

Lecture Notes

EC226 Term 1 2025/26

Dummy variables as explanatory variables

Readings:

- Stock and Watson (2003): Chapters 5.3 + 8.3
- Dougherty (2016): Chapter 5
- Wooldridge (2013): Chapter 7

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1 Additive dummy variables

Consider the following regression model:

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

we interpret the coefficient on the continuous variable x_1 as

$$\frac{\partial y_i}{\partial x_{1i}} = \beta_1 = \frac{\text{Change in } y_i}{\text{Unit increase in } x_1} \Big|_{\text{Holding all else constant}}$$

However, often our variables are qualitative in nature. For example, in an equation for wage we may want to include a variable for gender females appear to be paid less money for the same job, that is may want to specify a model:

$$\ln(w_i) = \alpha + \beta_1 School_i + \beta_2 F_i + \varepsilon_i \quad (1)$$

where, $School$ = Number of years of schooling, and $F = 1$ if female, 0 if non-female. The variable F is known as an additive dummy variable and has the effect of vertically shifting the regression line. To interpret the coefficient on the dummy variables it is best to write out the model for each category of the dummy variable relates. In our case the categorical variable is gender (which has two categories females and non-females). For non-females (when $F=0$), we have from (1):

$$E[\ln(w_i)] = \alpha + \beta_1 School_i \quad (2)$$

Whereas for females (when $F=1$), we have from (1)

$$\ln(w_i) = \alpha + \beta_1 School_i + \beta_2 + \varepsilon_i$$

that is,

$$[\ln(w_i)] = (\alpha + \beta_2) + \beta_1 School_i \quad (3)$$

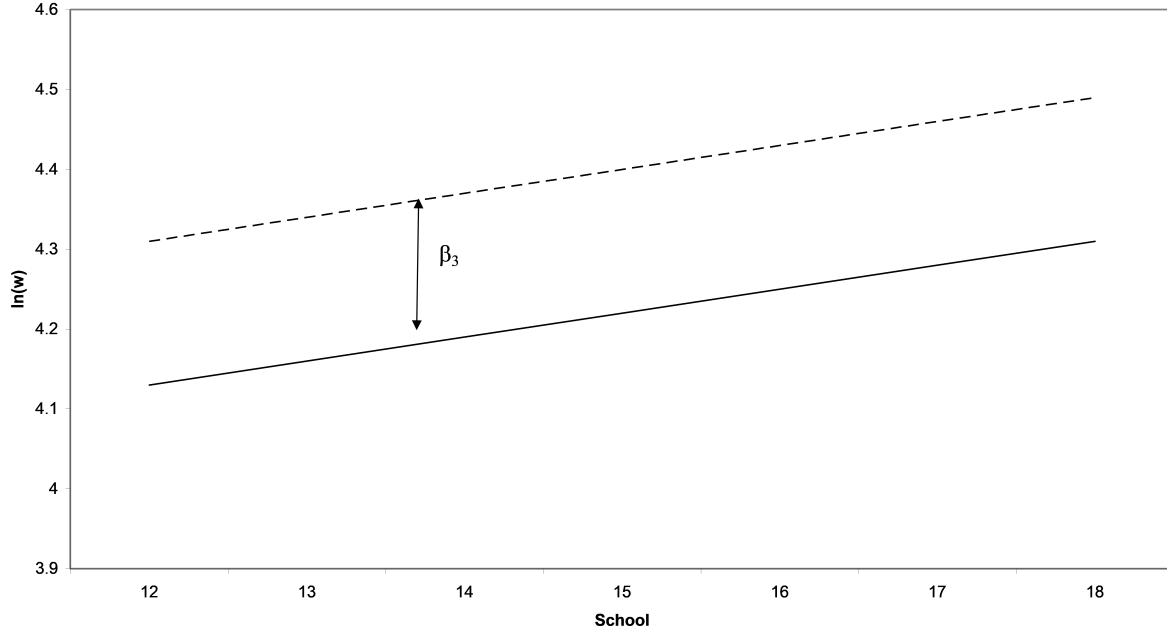
and so the intercept is α in equation (2) and is $(\alpha + \beta_2)$ in equation (3). So the two lines are parallel (as they have the same slope coefficient, β_1), but they cross the vertical axis at different points.

The difference between these two expressions for the same value of $School$ is:

$$E[\ln(w)^f] - E[\ln(w)^{nf}] = \beta_2$$

so β_2 is of the approximate expected proportionate change in the wages for a female compared to a non-female for a given value of *School* (assuming the coefficient is “small”). Pictorially this is shown in Figure 1, where the dashed line represents the regression line for non-females and the solid line that for females.

Figure 1: The role of an additive dummy variable



If instead of estimating equation (1), we estimated

$$\ln(w_i) = \delta_0 + \delta_1 School_i + \delta_2 NF_i + \varepsilon_i \quad (4)$$

where $NF = 1$ if non-female, 0 if female, as $NF = 1 - F$ then substituting into equation(4)

$$\begin{aligned} E[\ln(w_i)] &= \delta_0 + \delta_1 School_i + \delta_2(1 - F_i) + \varepsilon_i \\ E[\ln(w_i)] &= \delta_0 + \delta_1 School_i + \delta_2 - \delta_2 F_i + \varepsilon_i \\ E[\ln(w_i)] &= (\delta_0 + \delta_2) + \delta_1 School_i - \delta_2 F_i + \varepsilon_i \end{aligned} \quad (5)$$

Comparing equation (1) with equation (4) as the variables on the left and right hand side of the equation are identical the coefficients must be identical, in which case:

$$\delta_2 = -\beta_2, \quad \delta_1 = \beta_1, \quad \delta_0 = \alpha - \delta_2$$

and so by using NF in place of F in equation (1), all that happens is the coefficient on the NF dummy will be minus that on the F dummy and the coefficient on the

intercept will shift accordingly to allow for the fact that a Female is now the default case.

For the categorical variable gender (in which there are two categories), you include 1 dummy variable in the regression model (in our example F), so that non-female is the default and the coefficient on female is the change in the expected value of the dependent variable (for given values of the other variables) for females relative to non-females.

If we consider the categorical variable the type of industry in which people work; there may be are five broad categories (Manufacturing, Construction, Agriculture, Finance and Service), in which case you include four additive dummy variables, for example:

$$\ln(w_i) = \alpha + \beta_1 School_i + \beta_2 Cons_i + \beta_3 Agric_i + \beta_4 Serv_i + \beta_5 Fin_i + \varepsilon_i \quad (6)$$

Manufacturing the default industry type ($Cons_i = Agric_i = Serv_i = Fin_i = 0$)

$$E[\ln(w_i)] = \alpha + \beta_1 School_i$$

Construction ($Cons_i = 1, Agric_i = Serv_i = Fin_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_2) + \beta_1 School_i$$

Agriculture ($Agric_i = 1, Cons_i = Serv_i = Fin_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_3) + \beta_1 School_i$$

Service ($Serv_i = 1, Cons_i = Agric_i = Fin_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_4) + \beta_1 School_i$$

Finance ($Fin_i = 1, Cons_i = Agric_i = Serv_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_5) + \beta_1 School_i$$

So β_5 is the approximate proportionate change in the expected wages (assuming the coefficient is “small”) for somebody working in the Finance industry relative to Manufacturing Industry.

If there is a categorical variable with s categories, then you include $s - 1$ dummy variables and the omitted category is the default and the coefficient on any included category is then measured relative to the default (omitted) category.

2 Multiplicative dummy variables

In the equation for wages we may want to allow for the possibility that female pay, relative to non-female pay, also varies with the level of schooling:

$$E[\ln(w_i)] = \alpha + \beta_1 School_i + \beta_2 F_i + \beta_3 (F_i \times School_i) \quad (7)$$

where $School$ = Number of years of schooling and $F = 1$ if person is female, and 0 if non-female. The variable F is known as an additive dummy variable and $(F_i \times School_i)$ is known as a multiplicative dummy variable. The multiplicative dummy variable has the effect of rotating the regression line, so for non-female (when $F=0$) and from equation (7) we have:

$$E[\ln(w_i)] = \alpha + \beta_1 School_i \quad (8)$$

Whereas for females (when $F=1$) from equation (7) we have:

$$E[\ln(w_i)] = \alpha + \beta_1 School_i + \beta_2 + \beta_3 School_i$$

that is,

$$E[\ln(w_i)] = (\alpha + \beta_2) + (\beta_1 + \beta_3) School_i \quad (9)$$

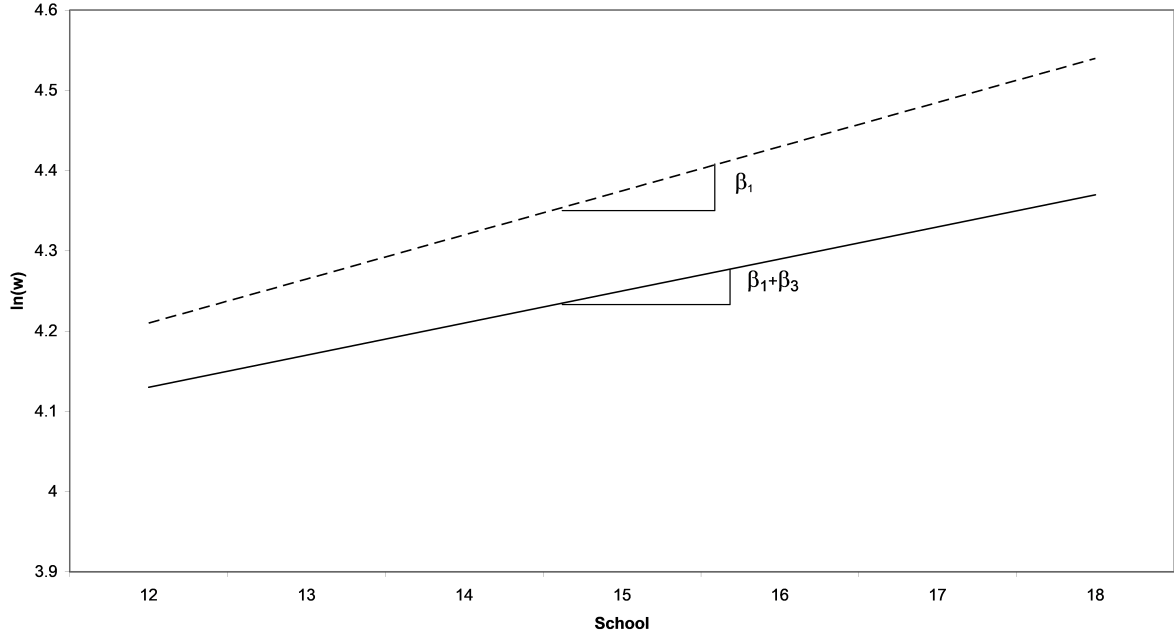
The intercept is α and the coefficient on $School$ is β_1 in equation (8) and the intercept is $(\alpha + \beta_2)$ and the coefficient on $School$ is $(\beta_1 + \beta_3)$ in equation (9). So β_2 is the approximate proportionate change in expected wages for a female compared to a non-female for $School=0$, (assuming the coefficient is “small”) while,

$$\left. \frac{\partial \ln(w_i)}{\partial School} = \beta_1 \right|_{\text{All else constant and non-female}}$$

$$\left. \frac{\partial \ln(w_i)}{\partial School} = (\beta_1 + \beta_3) \right|_{\text{All else constant and female}}$$

and β_3 is the increase in the semi-elasticity of wages with respect to an extra year of schooling for females compared to males. Pictorially this is shown in Figure 2, where the dashed line represents the regression line for males and the solid line that for females.

Figure 2: The role of an multiplicative dummy variable



As gender has two categories, you can only include 1 multiplicative dummy variable in our regression model ($F \times School$), so that non-female is the default and the coefficient on $F \times School$ is the change in the coefficient on School (for given values of the other variables) for females compared to non-females.

If we consider the categorical variable industry-type in which people work and there are five industry categories (Manufacturing, Construction, Agriculture, Finance and Service) in which case you could include four additive and four multiplicative dummy variables, for example:

$$E[\ln(w_i)] = \alpha + \beta_1 School_i + \beta_2 Cons_i + \beta_3 Agric_i + \beta_4 Serv_i + \beta_5 Fin_i + \beta_6 Cons \times School + \beta_7 Agric \times School + \beta_8 Serv \times School + \beta_9 Fin \times School \quad (10)$$

Manufacturing the default industry type ($Cons_i = Agric_i = Serv_i = Fin_i = 0$)

$$E[\ln(w_i)] = \alpha + \beta_1 School_i$$

Construction ($Cons_i = 1, Agric_i = Serv_i = Fin_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_2) + (\beta_1 + \beta_6) School_i$$

Agriculture ($Agric_i = 1, Cons_i = Serv_i = Fin_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_3) + (\beta_1 + \beta_7)School_i$$

Service ($Serv_i = 1, Cons_i = Agric_i = Fin_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_4) + (\beta_1 + \beta_8)School_i$$

Finance ($Fin_i = 1, Cons_i = Agric_i = Serv_i = 0$)

$$E[\ln(w_i)] = (\alpha + \beta_5) + (\beta_1 + \beta_9)School_i$$

So β_5 is the approximate proportionate change in expected wages (assuming the coefficient is “small”) for somebody working in the Finance industry relative to Manufacturing Industry (when $School = 0$) and β_9 is the change in the semi-elasticity of wages with respect to $School$ (for given values of the other variables) for somebody working in the Finance compared to Manufacturing industry.

If there is a categorical variable with s categories, then you include $s - 1$ multiplicative dummy variables (multiplied by $School$) and the omitted category is the default and the coefficient on any of the multiplicative dummy variables is measured relative to the default (omitted) category.

3 Interactive dummy variables

In the equation for wage we may also want to allow for the possibility that two dummy variables could be multiplied together, for example, to allow in the model females to be paid less than non-females and that the differential be different for non-whites (Nwh):

$$E[\ln(w_i)] = \beta_1 + \beta_2 F_i + \beta_3 Nwh_i + \beta_4 (F_i \times Nwh_i) \quad (11)$$

We have 4 alternative cases

1. Non-female-white (M, Wh)

$$\begin{aligned} E[\ln(w_i)] &= \beta_1 + \beta_2 \cdot 0 + \beta_3 \cdot 0 + \beta_4 \cdot 0 \cdot 0 \\ E[\ln(w_i)] &= \beta_1 \end{aligned} \quad (12)$$

2. Female-white (F, Wh)

$$\begin{aligned} E[\ln(w_i)] &= \beta_1 + \beta_2 \cdot 1 + \beta_3 \cdot 0 + \beta_4 \cdot 1 \cdot 0 \\ E[\ln(w_i)] &= \beta_1 + \beta_2 \end{aligned} \quad (13)$$

3. Non-female-Non-white ($NF, N - Wh$)

$$\begin{aligned} E[\ln(w_i)] &= \beta_1 + \beta_2 \cdot 0 + \beta_3 \cdot 1 + \beta_4 \cdot 0 \cdot 1 \\ E[\ln(w_i)] &= \beta_1 + \beta_3 \end{aligned} \quad (14)$$

4. Female-Non-white ($F, N - Wh$)

$$\begin{aligned} E[\ln(w_i)] &= \beta_1 + \beta_2 \cdot 1 + \beta_3 \cdot 1 + \beta_4 \cdot 1 \cdot 1 \\ E[\ln(w_i)] &= \beta_1 + \beta_2 + \beta_3 + \beta_4 \end{aligned} \quad (15)$$

Comparing equation (13) to equation (12):

$$E[\ln(w_i)^{F,Wh}] - E[\ln(w_i)^{M,Wh}] = \beta_2 \quad (16)$$

that is β_2 = Approximate proportionate change in expected wages for females compared to males, given they are white (assuming the coefficient is “small”).

Comparing equation (14) to equation (12):

$$E \left[\ln(w_i)^{NF, Non-Wh} \right] - E \left[\ln(w_i)^{M, Wh} \right] = \beta_3 \quad (17)$$

That is β_3 = Approximate proportionate change in expected wages for Whites compared to Non-whites, given they are non-females (assuming the coefficient is “small”).

Comparing equation (15) to equation (14):

$$E \left[\ln(w_i)^{F, Non-Wh} \right] - E \left[\ln(w_i)^{NF, Non-Wh} \right] = (\beta_2 + \beta_4) \quad (18)$$

that is $\beta_2 + \beta_4$ = Approximate proportionate change in expected wages for females compared to non-females, given they are non-white (assuming the sum of the two coefficient is “small”).

Comparing equation (18) to equation (16)

$$\left\{ E \left[\ln(w_i)^{F, Non-Wh} \right] - E \left[\ln(w_i)^{NF, Non-Wh} \right] \right\} - \left\{ E \left[\ln(w_i)^{F, Wh} \right] - E \left[\ln(w_i)^{NF, Wh} \right] \right\} = \beta_4 \quad (19)$$

That is β_4 = Additional proportionate change in expected wages for females (compared to non-females) who are non-white compared to that gender differential for those who are white.

	Male	Female	Gender effect	
White	β_1	$\beta_1 + \beta_2$	β_2	
Non-white	$\beta_1 + \beta_3$	$\beta_1 + \beta_2 + \beta_3 + \beta_4$	$\beta_2 + \beta_4$	β_4
Ethnicity effect	β_3	$\beta_3 + \beta_4$		
		β_4		

This is an example of a difference-in-difference model as the coefficient on the interaction term is interpreted as the difference between two differences (see equation (19)).

Suppose we know that average hourly wages listed below:

	Male	Female	Gender effect	
White	15.09	12.58	-2.51	
Non-white	15.29	12.50	-2.79	-0.28
Ethnicity effect	+0.20	-0.08		
		-0.28		

Then using these numbers and estimating equation (8) by OLS would yield: $b_1 = 15.09$, $b_2 = -2.51$, $b_3 = +0.20$ and $b_4 = -0.28$.

Appendix A has an example of dummy variables from Stata, whereas Appendix B has a worked example on dummy variables.

Appendices

A Stata dummy variable regression

```
regress lhourpay age c.age#c.age edage i.ethnic if(edage>=0 & edage<=95)
```

Source	SS	df	MS	Number of obs	=	9,526
Model	429.751111	5	85.9502222	F(5, 9520)	=	321.73
Residual	2543.2957	9,520	.26715291	Prob > F	=	0.0000
				R-squared	=	0.1445
				Adj R-squared	=	0.1441
Total	2973.04682	9,525	.312130899	Root MSE	=	.51687

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0612652	.0027281	22.46	0.000	.0559176	.0666128
c.age#c.age	-.0006197	.0000311	-19.95	0.000	-.0006806	-.0005588
edage	.0528746	.0017666	29.93	0.000	.0494116	.0563376
ethnic						
Asian	-.137411	.0240132	-5.72	0.000	-.1844819	-.0903401
Other	-.1078589	.0255227	-4.23	0.000	-.1578889	-.057829
_cons	.1792843	.0637394	2.81	0.005	.0543414	.3042271

So compared to whites, Asian individuals earn on average approximately 0.137 proportionately (or 13.7%) [exactly this is 12.8%] less than white individuals and individuals from other ethnic groups on average earn approximately 0.108 proportionately (10.8%) less than white individuals.

```
regress lhourpay age c.age#c.age edage i.ethnic i.ethnic#c.edage if(edage>=0 & edage<=95)
```

Source	SS	df	MS	Number of obs	=	9,526
Model	437.095708	7	62.4422441	F(7, 9518)	=	234.36
Residual	2535.95111	9,518	.266437393	Prob > F	=	0.0000
				R-squared	=	0.1470
				Adj R-squared	=	0.1464
Total	2973.04682	9,525	.312130899	Root MSE	=	.51618

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.061255	.0027251	22.48	0.000	.0559133	.0665967
c.age#c.age	-.0006185	.000031	-19.93	0.000	-.0006793	-.0005576
edage	.055195	.0019117	28.87	0.000	.0514477	.0589424
ethnic						
Asian	-.2841772	.1465	-1.94	0.052	-.5713486	.0029941
Other	.5160296	.1256464	4.11	0.000	.269736	.7623233
ethnic#c.edage						
Asian	.0070374	.0071606	0.98	0.326	-.0069989	.0210738
Other	-.0311351	.0061496	-5.06	0.000	-.0431897	-.0190805
_cons	.1343662	.0649992	2.07	0.039	.0069539	.2617785

We find that there is a much lower return to education for individuals from other ethnic groups compared to white individuals. The return to education for whites is 0.0552 (approximately 5.52% per year), whereas for other ethnic groups it is only 0.0552-0.0311=0.0241 (approximately 2.41% per year).

B Exercise questions

A model for the transfer price of all footballers sold by a Premier League team in England and Wales in the year 2010/11 is written as

$$\ln(p_i) = \alpha + \beta_1 \ln(\text{Age}) + \beta_2 \ln(\text{Age})^2 + \beta_3 \ln(\text{App}_i) + \beta_4 \text{Str}_i + \beta_5 \text{For}_i + \varepsilon_i \quad (\text{A1})$$

where p = Transfer price of each player, Age = Age of the player, App = Number of first team appearances the player has made, $\text{For} = 1$ if the player is a non-UK player, zero otherwise, $\text{Str} = 1$ if the player is a striker, zero for a non-striker.

Using data for 154 transfers of players equation (A1) is estimated by OLS and the results are:

$$\ln(p_i) = \underset{(1.71)}{3.426} + \underset{(0.012)}{0.024} \ln(\text{Age}_i) - \underset{(0.006)}{0.008} \ln(\text{Age}_i)^2 + \underset{(0.099)}{0.212} \ln(\text{App}_i) + \underset{(0.092)}{0.216} \text{Str}_i - \underset{0.076}{0.312} \text{For}_i + e_i$$

$$RSS = 1.932, R^2 = 0.293 \text{ (Standard errors are in parentheses)}$$

- (a) Briefly interpret the coefficients.
- (b) Test the hypotheses (i) $\beta_3 = 0$, (ii) the elasticity of price with respect to age is zero for a player aged 25, given $\text{cov}(b_1, b_2) = -0.00009$.
- (c) Test the hypothesis that all slope coefficients are jointly equal to zero.
- (d) The interactive dummy variable ($\text{Str} \times \text{For}$) is added into equation (1). Interpret the coefficients on the three dummy variables in this new model.
- (e) There are 20 clubs in the Premier League. Including 19 dummy variables into equation (A1) for each club (except Leicester City) yielded an RSS of 1.844. Interpret the coefficients on one of these 19 dummy variables and test for their joint significance.

Answer

(a) In interpreting the coefficients:

$$\frac{\partial \ln(p)}{\partial \ln(Age)} = \beta_1 + 2\beta_2 \ln(Age) = \text{Elasticity of price w.r.t age}$$

$$\frac{\partial \ln(p)}{\partial \ln(App)} = \beta_3 = \text{Elasticity of price w.r.t appearance}$$

$$\frac{\partial \ln(p)}{\partial Str} = \beta_4 = \text{Proportionate increase in price for a striker compared to a non – striker}$$

$$\frac{\partial \ln(p)}{\partial For} = \beta_5 = \text{Proportionate increase in price for a foreign compared to UK player}$$

(b) (i) $H_0 : \beta_3 = 0; H_1 : \beta_3 \neq 0$ $t_{154-6}^{0.05} = 1.645$

$$t = \frac{0.212 - 0}{0.099} = 2.141 \Rightarrow \text{Reject } H_0$$

(ii) $H_0 : \beta_1 + 2\beta_2 \ln(25) = 0 \Rightarrow H_0 : \beta_1 + 6.44\beta_2 = 0$

$$t = \frac{0.024 - 6.44(0.008) - 0}{\sqrt{0.012^2 + 6.44^2(0.006)^2 + 12.88(-0.00009)}} = \frac{-0.02752}{\sqrt{0.0004778}} = 1.259 \Rightarrow \text{Do not reject } H_0$$

(c) $H_0 : \beta_1 = \beta_2 = \dots \beta_5 = 0; H_1 : \beta_j \neq 0$ for any j $F_{5,148}^{0.05} = 2.275$

$$F = \frac{0.293/5}{(1 - 0.293)/148} = 12.26 \Rightarrow \text{Reject } H_0$$

(d) Adding an interactive term into the equation (A1) our equation is:

$$\ln(p_i) = \alpha + \beta_1 \ln(Age) + \beta_2 \ln(Age)^2 + \beta_3 \ln(App_i) + \beta_4 Str_i + \beta_5 For_i + \beta_6 (Str_i \times For_i) + \varepsilon_i$$

β_4 = Proportionate increase in price for a striker compared to a non-striker, given they are a non-foreign player.

β_5 = Proportionate increase in price for a foreign player versus a UK player, given they are a non-striker).

$$\beta_6 = (E[\ln(p_i)|_{Str,For}] - E[\ln(p_i)|_{Str,UK}]) - (E[\ln(p_i)|_{NStr,For}] - E[\ln(p_i)|_{NStr,UK}])$$

(e) $H_0 : \gamma_1 = \gamma_2 = \dots \gamma_{19} = 0; H_0 : \gamma_j \neq 0$ for any j $F_{19,129}^{0.05} = 1.668$

$$F = \frac{(1.932 - 1.844)/19}{1.844/(154 - 25)} = 0.324 \Rightarrow \text{Do not reject } H_0$$

C Stata dummy variable commands

```

/* Recoding ETHEUKEUL with 9 categories into ethnic with 3 categories */
gen ethnic=1 if(ETHUKEUL==1)
replace ethnic=2 if(ETHUKEUL==3 | ETHUKEUL==4 | ETHUKEUL==5 | ETHUKEUL==6 /*
*/ | ETHUKEUL==7)
replace ethnic=3 if(ETHUKEUL==2 | ETHUKEUL==8 | ETHUKEUL==9)
label define ethnic 1 "White" 2 "Asian" 3 "Other"
label value ethnic ethnic
tab ethnic
/* Using the factor option to input the dummy variables form a categorical
variable the default is ethnicity with the lowest value and storing the
output as m1 */
reg lhourpay age c.age#c.age edage i.ethnic if(edage>=0 & edage<=95)
est store m1
/* testing the significance of the ethnic variables */
test 2.ethnic 3.ethnic
testparm i.ethnic
/* Changing the default category in ethnic and storing the output as m2 */
reg lhourpay age c.age#c.age edage ib3.ethnic if(edage>=0 & edage<=95)
est store m2
/* Including a multiplicative term between ethnicity and edage and storing
output as m3 */
reg lhourpay age c.age#c.age edage i.ethnic i.ethnic#c.edage if(edage>=0 /*
*/ & edage<=95)
est store m3
/* Producing the output as an excel file multi_reg */
outreg2 [m1 m2 m3] using multi_reg, replace excel
/* Testing the significance of the mutiplicative terms */
test 2.ethnic#c.edage 3.ethnic#c.edage
testparm i.ethnic#c.edage
/* Changing the default of ethnicity */
reg lhourpay age c.age#c.age edage ib3.ethnic ib3.ethnic#c.edage if(edage>=0 /*
*/ & edage<=95)
/* Using ## option to include i.ethnic, edage and the mutiplication of the two */
reg lhourpay age c.age#c.age i.ethnic##c.edage if(edage>=0 & edage<=95)
/* Testing the significance of i.ethnic, edage and the multiplcation of the two */
testparm i.ethnic##c.edage

```

```
/* Including an interaction term between ethnicity and gender */
reg lhourpay age c.age#c.age i.SEX i.ethnic if(edage>=0 & edage<=95)
/* Changing the defaults of gender and ethnicity */
reg lhourpay age c.age#c.age ib2.SEX ib3.ethnic if(edage>=0 & edage<=95)
/* Including an interaction term between gender and ethnicity */
reg lhourpay age c.age#c.age i.SEX i.ethnic i.sex#i.ethnic if(edage>=0 & /*
*/ edage<=95)
/* An equivalent regression would be */
reg lhourpay age c.age#c.age i.SEX##i.ethnic if(edage>=0 & edage<=95)
/* A 3-way interaction term including part-time as well would be */
gen ftpt=(FTPT==1)
replace ftpt=. if(FTPT<0 | FTPT>2)
reg lhourpay age c.age#c.age i.SEX##i.ethnic##i.ftpt if(edage>=0 & edage<=95)
```

D R dummy variable commands

#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:

```
library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)

qlfs18 <- qlfs18 %>%
  mutate(ETHUKEUL1=as.factor(ETHUKEUL))
qlfs18 <- qlfs18 %>%
  mutate(ethnic= ifelse(ETHUKEUL1==1, 1,
    ifelse(ETHUKEUL1==3 | ETHUKEUL1==4 | ETHUKEUL1==5 | ETHUKEUL1==6 |
    ETHUKEUL1==7, 2,
    ifelse(ETHUKEUL1==2 | ETHUKEUL1==8 | ETHUKEUL1==9, 3, NA))))
qlfs18$ethnic <- factor(qlfs18$ethnic, levels = c(1, 2, 3),
  labels = c("White", "Asian", "Other"))
# Using the factor option to input the dummy variables form a categorical variable
#the default is ethnicity with the lowest value
model5 <- lm(lhourpay ~ age + I(age^2) + edage1 + ethnic,
  data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model5)
```

```
# testing the significance of the ethnic variables
test5_1 <- linearHypothesis(model5, c("ethnicAsian = 0", "ethnicOther=0"),
                             test="F")
fstat5_1 <- test5_1$F[2]
# Changing the default category in ethnic
model6 <- lm(lhourpay ~ age + I(age^2) + edage1 + relevel(ethnic, ref = 3),
             data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model6)
# including a multiplicative term between ethnicity and edage
model7 <- lm(lhourpay ~ age + I(age^2) + edage1 + ethnic + ethnic:edage,
             data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model7)
# testing the significance of the multiplicative terms
test7_1 <- linearHypothesis(model7, c("edage:ethnicAsian = 0",
                                     "edage:ethnicOther=0"), test="F")
fstat7_1 <- test7_1$F[2]
# Changing the default of ethnicity
model8 <- lm(lhourpay ~ age + I(age^2) + edage1 + relevel(ethnic, ref = 3) +
             relevel(ethnic, ref = 3):edage, data = subset(qlfs18,
             edage1>=0 & edage1<=95))
summary(model8)
# Including an interaction term between ethnicity and gender
qlfs18<- qlfs18 %>%
  mutate(sex = as.factor(SEX),
         )
qlfs18$sex <- factor(qlfs18$sex, levels = c(1, 2), labels = c("Male", "Female"))

model9 <- lm(lhourpay ~ age + I(age^2) + sex + ethnic,
             data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model9)
# Changing the defaults of gender and ethnicity */
model10 <- lm(lhourpay ~ age + I(age^2) + relevel(sex, ref=2) +
             relevel(ethnic, ref=3),
             data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model10)
# Including an interaction term between gender and ethnic
model11 <- lm(lhourpay ~ age + I(age^2) + sex + ethnic + sex:ethnic,
```

```
data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model11)
# An equivalent regression would be
model12 <- lm(lhourpay ~ age + I(age^2) + sex*ethnic,
             data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model12)
# A 3-way interaction term including part-time as well would be
qlfs18<- qlfs18 %>%
  mutate(FTPT = as.factor(FTPT),
         ftpt = ifelse(FTPT==1, 1, 0))
qlfs18$ftpt <- factor(qlfs18$ftpt, levels = c(1, 0),
                    labels = c("full-time", "part-time"))
model13 <- lm(lhourpay ~ age + I(age^2) + sex + ethnic + ftpt + sex:ethnic
             + sex:ftpt + ethnic:ftpt + sex:ethnic:ftpt,
             data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model13)
# An alternative 3-way interaction term model
model14 <- lm(lhourpay ~ age + I(age^2) + sex * ethnic * ftpt,
             data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model14)
# Creating a nice table of results
stargazer(model7, model8, model9, type='text', digits = 2,
          title = 'Table 1: Some Regression Results',
          style = 'qje',
          column.labels = c('Model 7', 'Model 8', 'Model 9'),
          dep.var.labels = 'ln(Hourpay)')
```

References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.